Submission B solution to Problem 8

Overview and scope

The original question asks whether an order-reversing involution on the flat lattice of a matroid extends to an order-reversing involution of the relative Dressian. The theorem proved here is the finite-ground-set version. The proof first reduces to the simplification of the matroid. In the simple case, the existence of a flat-lattice anti-involution forces the lattice of flats to be modular. Modularity then implies that every valuated quotient is constant on the bases of each flat, which permits the construction of a dual, or "dagger," valuated quotient.

The main work is to prove that the dagger construction is again a valuated quotient, that it is involutive, and that it depends only on the tropical linear space rather than on the chosen valuated Pluecker vector. The first point is proved by checking the support matroid, the tropical Pluecker relations, and the quotient circuit criterion. The second is formal from the definition. The third is an annihilator statement using polar bases. Finally the construction is transported back from the simplification to the original finite matroid.

Theorem 1 (Established finite-ground-set theorem)**.** *Let M be a finite matroid on ground set E. Suppose that the lattice of flats $\mathcal{L}(M)$ admits an order-reversing involution*

$$F \longmapsto F^{\perp}.$$

Let $\mathrm{Dr}(M)$ *be the poset, ordered by inclusion, of projective tropical linear spaces contained in* $\mathrm{Trop}(M)$*. Then there is an order-reversing involution*

$$\Phi : \mathrm{Dr}(M) \longrightarrow \mathrm{Dr}(M)$$

such that, for every flat $F \in \mathcal{L}(M)$,

$$\Phi(\mathrm{Trop}(M/F \oplus U_{0,F})) = \mathrm{Trop}(M/F^{\perp} \oplus U_{0,F^{\perp}}),$$

where $U_{0,F}$ denotes the rank-zero matroid on F, and the direct sums are considered as matroids on E.

Preliminaries

We use the min-plus semiring $\mathbb{T} = \mathbb{R} \cup \{\infty\}$. Tropical projective space is obtained from $\mathbb{T}^E \setminus \{(\infty, \dots, \infty)\}$ by quotienting by simultaneous tropical scalar translation; in affine-cone calculations we allow the all-∞ vector.

A rank k valuated matroid on E is a function

$$p : \binom{E}{k} \to \mathbb{T}$$

not identically ∞, satisfying the tropical Pluecker relations; we extend p to all subsets by $p_I = \infty$ when $|I| \neq k$. Its tropical linear space is

$$\mathrm{Trop}(p) = \left\{ x \in \mathbb{TP}^{E-1} : \text{ for all } J \in \binom{E}{k+1},\ \min_{j \in J}(x_j + p_{J-j}) \text{ has no unique finite minimizer} \right\}.$$

An ordinary matroid M of rank r is identified with the constant valuated matroid p^M, where $p^M_B = 0$ for bases B of M, and $p^M_B = \infty$ otherwise. A valuated matroid p is called a valuated quotient of M if $\mathrm{Trop}(p) \subseteq \mathrm{Trop}(M)$. Thus $\mathrm{Dr}(M)$ is the set of all projective spaces $\mathrm{Trop}(p)$, with p a valuated quotient of M, ordered by inclusion.

We use standard facts from valuated matroid theory: the support of a valuated matroid is an ordinary matroid; support plus the three-term tropical Pluecker relations characterize valuated matroids; tropical linear spaces are the tropical convex hulls of their valuated cocircuits; and tropical hyperplanes are tropically convex. These are due to Dress–Wenzel and Murota–Tamura and are used in the standard treatments by Speyer and by Maclagan–Sturmfels; see also Hampe for tropical convexity [1, 2, 3, 4, 5]. For flats of M, we write

$$G \wedge H = G \cap H, \qquad G \vee H = \mathrm{cl}_M(G \cup H).$$

Lemma 2 (Quotient criterion and support consequences)**.** *Let p be a rank k valuated matroid on E, and for $A \in \binom{E}{k-1}$ put $C_A(e) = p_{A \cup e}$.*

1. *If $k > 0$, then $\mathrm{Trop}(p) \subseteq \mathrm{Trop}(M)$ if and only if, for every circuit C of M and every $A \in \binom{E}{k-1}$,*

$$\min_{c \in C \setminus A} p_{A \cup c}$$

has no unique finite minimizer. If $k = 0$, then $\mathrm{Trop}(p)$ is empty projectively and p is automatically a quotient.

2. *If p is a quotient and $N = \mathrm{supp}(p)$, then:*

$$I \text{ dependent in } M,\ |I| = k \Longrightarrow p_I = \infty;$$

$$\mathrm{rk}_N(A) = \mathrm{rk}_N(\mathrm{cl}_M(A)) \quad \text{for all } A \subseteq E;$$

and for flats $G \subseteq H$ of M,

$$\mathrm{rk}_N(H) - \mathrm{rk}_N(G) \leq \mathrm{rk}_M(H) - \mathrm{rk}_M(G).$$

Proof. The Bergman fan $\mathrm{Trop}(M)$ is cut out by the circuit hyperplanes

$$\min_{c\in C} x_c \quad \text{has no unique finite minimizer}$$

for circuits C of M. Also $\mathrm{Trop}(p)$ is the tropical convex hull of the valuated cocircuits C_A, and tropical hyperplanes are tropically convex. Hence $\mathrm{Trop}(p) \subseteq \mathrm{Trop}(M)$ exactly when every C_A satisfies every circuit equation of M, which is the displayed criterion. For $k = 0$, the rank-zero tropical linear space has only the all-∞ affine vector, hence no projective point.

Now assume p is a quotient. If $I \in \binom{E}{k}$ is M-dependent, choose an M-circuit $C \subseteq I$ and $c \in C$. With $A = I - c$, the criterion gives a one-term minimum p_I; therefore $p_I = \infty$.

We next prove $\mathrm{rk}_N(A) = \mathrm{rk}_N(\mathrm{cl}_M(A))$. It is enough to show $\mathrm{cl}_M(A) \subseteq \mathrm{cl}_N(A)$. Let C be an M-circuit. If some $e \in C$ were a coloop of $N|C$, then there would be an N-hyperplane H containing $C - e$ but not e. Choose an N-basis A of H. Then $A \cup e$ is an N-basis while $A \cup c$ is N-dependent for every $c \in C - e$, contradicting the quotient criterion. Thus no element of C is a coloop of $N|C$, so every element of C lies in an N-circuit contained in C. If $e \in \mathrm{cl}_M(A)$, choose an M-circuit $C \subseteq A \cup e$ containing e; an N-circuit in C containing e proves $e \in \mathrm{cl}_N(A)$.

Finally, if $G \subseteq H$ are M-flats, choose $S \subseteq H$ with $\mathrm{cl}_M(G \cup S) = H$ and $|S| = \mathrm{rk}_M(H) - \mathrm{rk}_M(G)$. Then

$$\mathrm{rk}_N(H) = \mathrm{rk}_N(G \cup S) \le \mathrm{rk}_N(G) + |S|,$$

which is the desired inequality. □

Lemma 3 (Reduction to the simplification)**.** *Let $L_0 = \mathrm{cl}_M(\varnothing)$ be the loop set of M, let $S = \mathrm{si}(M)$, let $\bar{E}$ be the set of nonloop parallel classes, and let*

$$\pi : E \setminus L_0 \to \bar{E}$$

be the quotient map. Define

$$(\Delta x)_e = \begin{cases} \infty, & e \in L_0, \\ x_{\pi(e)}, & e \notin L_0. \end{cases}$$

Then Δ induces a poset isomorphism $\mathrm{Dr}(S) \cong \mathrm{Dr}(M)$. Moreover, if $F \in \mathcal{L}(M)$ corresponds to $\bar{F} \in \mathcal{L}(S)$, then

$$\mathrm{Trop}(M/F \oplus U_{0,F}) = \Delta\big(\mathrm{Trop}(S/\bar{F} \oplus U_{0,\bar{F}})\big).$$

Proof. Every flat of M contains L_0 and is a union of nonloop parallel classes, so $\mathcal{L}(M) \cong \mathcal{L}(S)$.

Let p be a rank k quotient of M. By Lemma 2, every M-dependent k-set has value ∞. Hence no finite basis contains a loop or two parallel elements. If e, f are parallel nonloops and $A \cap \{e, f\} = \varnothing$, the circuit $\{e, f\}$ gives $p_{A\cup e} = p_{A\cup f}$. Therefore p descends uniquely to a valuated matroid $\bar{p}$ on $\bar{E}$. The tropical Pluecker relations and the quotient circuit criterion for $\bar{p}$ are obtained from those for p, because circuits of S are precisely images of nonloop circuits of M after deleting parallel repetitions. Thus $\bar{p}$ is a quotient of S.

Conversely, if q is a quotient of S, define

$$q^{\uparrow}_I = \begin{cases} q_{\pi(I)}, & I \cap L_0 = \varnothing \text{ and } \pi \text{ is injective on } I, \\ \infty, & \text{otherwise.} \end{cases}$$

The Pluecker relations for $q^{\uparrow}$ either reduce under π to those for q, or are trivial because all finite terms occur in equal parallel pairs. The quotient criterion for M follows from the quotient criterion for S, together with the loop circuits and two-element parallel circuits. Thus $q^{\uparrow}$ is a quotient of M.

The tropical equations of $q^{\uparrow}$ force loop coordinates to be ∞ and force equality along each nonloop parallel class; after this identification the remaining equations are exactly those of q. Hence $\mathrm{Trop}(q^{\uparrow}) = \Delta(\mathrm{Trop}(q))$, and inclusions are preserved and reflected. This proves the poset isomorphism. The final displayed identity is the same statement applied to the ordinary quotient $M/F \oplus U_{0,F}$, whose simplification is $S/\bar{F} \oplus U_{0,\bar{F}}$. □

The simple case

Assume in this section that M is simple of rank r, and that $\mathcal{L}(M)$ has the given order-reversing involution.

Lemma 4 (Modularity and flat constancy)**.** *The matroid M is modular:*

$$\mathrm{rk}_M(G) + \mathrm{rk}_M(H) = \mathrm{rk}_M(G \wedge H) + \mathrm{rk}_M(G \vee H)$$

for all flats G, H. Also $\mathrm{rk}_M(F^{\perp}) = r - \mathrm{rk}_M(F)$. If p is a rank k valuated quotient of M, then $p_I = p_{I'}$ whenever I, I' are M-bases of the same rank k flat.

Proof. Since M is simple, $\mathcal{L}(M)$ is a finite geometric lattice, hence semimodular. The anti-involution sends maximal chains from $\varnothing$ to F to maximal chains from $F^{\perp}$ to E, so $\mathrm{rk}(F^{\perp}) = r - \mathrm{rk}(F)$. Applying semimodularity to G, H gives "$\geq$" in the modular identity. Applying semimodularity to $G^{\perp}, H^{\perp}$ and using rank reversal gives the opposite inequality, hence equality.

For flat constancy, the basis graph of $M|F$ is connected, so it suffices to compare $I = A \cup a$ and $I' = A \cup b$. Since $\mathrm{cl}(Aa) = \mathrm{cl}(Ab) = F$, there is an M-circuit $C \subseteq A \cup \{a, b\}$ containing both a and b. Lemma 2 applied to C and A says the minimum of p_{Aa} and p_{Ab} has no unique finite minimizer; therefore $p_{Aa} = p_{Ab}$, allowing both sides to be ∞. □

For a rank k quotient p, write

$$\bar{p}(F) = p_I$$

when F is a rank k flat and I is any M-basis of F. This is well-defined by Lemma 4. Put $n = r - k$, and define

$$p^{\dagger}_J = \begin{cases} \bar{p}(\mathrm{cl}_M(J)^{\perp}), & |J| = n \text{ and } J \text{ is } M\text{-independent}, \\ \infty, & |J| = n \text{ and } J \text{ is } M\text{-dependent.} \end{cases}$$

Lemma 5 (Local dual Pluecker relation)**.** *Let p be a rank k quotient of M. Let Y be a rank $k+2$ flat and let H_1, H_2, H_3, H_4 be hyperplanes of Y. Define*

$$[ij] = \begin{cases} \bar{p}(H_i \wedge H_j), & H_i \neq H_j, \\ \infty, & H_i = H_j. \end{cases}$$

Then

$$\min\bigl([12] + [34],\ [13] + [24],\ [14] + [23]\bigr)$$

has no unique finite minimizer.

Proof. If two hyperplanes are equal, say $H_1 = H_2$, then $[12] = \infty$, while $[13] = [23]$ and $[14] = [24]$; hence the last two sums are equal. Assume now that the H_i are distinct, and put $Z = H_1 \wedge H_2 \wedge H_3 \wedge H_4$. In the modular interval $[\varnothing, Y]$, $\mathrm{rk}(Z) \in \{k-2, k-1, k\}$.

If $\mathrm{rk}(Z) = k$, every pairwise meet is Z, so all three sums are equal. If $\mathrm{rk}(Z) = k-2$, work in the rank-four interval $[Z, Y]$. Let $T_i = \bigwedge_{j \neq i} H_j$. Modularity gives $\mathrm{rk}(T_i/Z) = 1$: it is at least 1, and if it were at least 2, then $T_i \wedge H_i$ would have positive relative rank, contradicting $T_i \wedge H_i = Z$. For $\{i, j, m, n\} = \{1, 2, 3, 4\}$, one has $H_i \wedge H_j = T_m \vee T_n$. Choose a basis A of Z and $t_i \in T_i \setminus Z$. Then, for example, $[12] = p_{At_3t_4}$ and $[34] = p_{At_1t_2}$, and the desired relation is the three-term tropical Pluecker relation for p on A, t_1, t_2, t_3, t_4.

It remains to treat $\mathrm{rk}(Z) = k-1$. In the rank-three modular geometry $[Z, Y]$, define $w(P) = \bar{p}(P)$ for points P. We first show $w \in \mathrm{Trop}([Z, Y])$. Let D be a circuit of this geometry, choose a basis A of Z, and choose $e_P \in P \setminus Z$ for $P \in D$. A circuit of the contraction by Z lifts to an M-circuit $C \subseteq Z \cup \{e_P : P \in D\}$ whose image is D. Applying Lemma 2 to C and A, the elements in Z contribute ∞, and the remaining values are $w(P)$. Thus $w \in \mathrm{Trop}([Z, Y])$.

We use the following elementary rank-three fact. If G is a finite simple modular rank-three matroid, $w \in \mathrm{Trop}(G)$, and $L_1, \ldots, L_4$ are distinct lines, then

$$\min_{\text{matchings}} \bigl(w(L_i \wedge L_j) + w(L_m \wedge L_n)\bigr)$$

has no unique finite minimizer. Indeed, for the min convention, $w \in \mathrm{Trop}(G)$ iff every superlevel set $\{x : w_x \geq \lambda\}$ is a flat: a nonclosed superlevel set gives a circuit with a unique minimum outside it, and conversely a unique circuit minimum gives such a superlevel set. In rank three the nontrivial superlevel flats form a chain $a \leq B$, where a is a point and B a line; after subtracting the minimum,

$$w(x) = \lambda \mathbf{1}_{x=a} + \mu \mathbf{1}_{x \leq B}, \qquad \lambda, \mu \in [0, \infty].$$

If none of the L_i equals B, color L_i by $y_i = L_i \wedge B$. A pair L_i, L_j meets on B exactly when $y_i = y_j$; otherwise its contribution is 0. For four colored objects, by the multiplicities $1+1+1+1, 2+1+1, 2+2, 3+1, 4$, the minimum over the three perfect matchings is never unique. If, say, $L_1 = B$, put $y_i = L_i \wedge B$ for $i = 2, 3, 4$ and $c(y) = \lambda + \mu$ for $y = a$, $c(y) = \mu$ otherwise. The three costs are

$$c(y_2) + \mathbf{1}_{y_3=y_4} c(y_3), \quad c(y_3) + \mathbf{1}_{y_2=y_4} c(y_2), \quad c(y_4) + \mathbf{1}_{y_2=y_3} c(y_2),$$

and the cases "all distinct", "exactly two equal", and "all equal" give, respectively, at least two minimal costs, the two costs pairing B with one of the equal colors, or three equal costs.

Apply this fact to the lines H_i/Z in $[Z, Y]$. Their pairwise meets are the flats $H_i \wedge H_j$, so the desired relation follows. $\square$

Lemma 6 (The dagger construction is a valuated quotient)**.** *For every rank k valuated quotient p of the simple modular matroid M, the function $p^\dagger$ is a rank $r-k$ valuated quotient of M.*

Proof. The cases $k = 0$ and $k = r$ are immediate: $p^\dagger$ is respectively the constant valuated matroid of M, or the unique rank-zero valuated matroid, up to adding a scalar. Assume $0 < k < r$, put $n = r - k$, and let $N = \operatorname{supp}(p)$.

First determine the support. For a flat G, set

$$\rho^\dagger(G) = \operatorname{rk}_M(G) - k + \operatorname{rk}_N(G^\perp),$$

and extend by $\rho^\dagger(A) = \rho^\dagger(\operatorname{cl}_M(A))$. We have $\rho^\dagger(\varnothing) = 0$ and $\rho^\dagger(E) = n$. If $G \subseteq H$ are flats, then $H^\perp \subseteq G^\perp$, and Lemma 2 gives

$$\operatorname{rk}_N(G^\perp) - \operatorname{rk}_N(H^\perp) \leq \operatorname{rk}_M(G^\perp) - \operatorname{rk}_M(H^\perp) = \operatorname{rk}_M(H) - \operatorname{rk}_M(G),$$

so $\rho^\dagger$ is monotone. For flats G, H, modularity of M, submodularity of rk_N, and the quotient property $\operatorname{rk}_N(A) = \operatorname{rk}_N(\operatorname{cl}_M A)$ give

$$\rho^\dagger(G) + \rho^\dagger(H) \geq \rho^\dagger(G \wedge H) + \rho^\dagger(G \vee H).$$

For arbitrary A, B, apply this to $G = \operatorname{cl}_M(A)$, $H = \operatorname{cl}_M(B)$, using $\operatorname{cl}(A \cup B) = G \vee H$ and $\operatorname{cl}(A \cap B) \subseteq G \wedge H$. Thus $\rho^\dagger$ is submodular. Since

$$0 \leq \rho^\dagger(A) \leq \operatorname{rk}_M(A) \leq |A|,$$

and submodularity gives $\rho^\dagger(A \cup e) - \rho^\dagger(A) \leq \rho^\dagger(\{e\}) \leq 1$, this is a matroid rank function of rank n. Call the matroid $N^\dagger$.

For $J \in \binom{E}{n}$, the condition $\rho^\dagger(J) = n$ forces J to be M-independent. For such J,

$$\rho^\dagger(J) = n - k + \operatorname{rk}_N(\operatorname{cl}_M(J)^\perp),$$

so J is a basis of $N^\dagger$ iff $\operatorname{rk}_N(\operatorname{cl}_M(J)^\perp) = k$, equivalently iff $p^\dagger_J < \infty$. Hence the finite support of $p^\dagger$ is a matroid.

It remains to check the three-term tropical Pluecker relations. Let $S \in \binom{E}{n-2}$ and let $a, b, c, d \notin S$ be distinct. If S is dependent, or if some of a, b, c, d lies in $X = \operatorname{cl}_M(S)$, all three Pluecker sums are ∞. Otherwise X has rank $n - 2$, $Y = X^\perp$ has rank $k + 2$, and

$$H_e = (X \vee e)^\perp \qquad (e = a, b, c, d)$$

is a hyperplane of Y. If $H_e = H_f$, then Sef is dependent and $p^\dagger_{Sef} = \infty$; if $H_e \neq H_f$, then Sef is independent and

$$p^\dagger_{Sef} = \bar{p}(H_e \wedge H_f).$$

Lemma 5 gives the required Pluecker relation. Thus $p^\dagger$ is a valuated matroid.

Finally we prove that it is a quotient of M. By Lemma 2, let C be an M-circuit and $A \in \binom{E}{n-1}$. If A is dependent, all terms $p^\dagger_{A\cup c}$ are ∞. Assume A is independent, put $X = \mathrm{cl}_M(A)$, and $Y = X^\perp$. For a point P of the simplification of M/X, equivalently a rank n flat P with $X < P$, define

$$u(P) = \bar{p}(P^\perp).$$

We claim $u \in \mathrm{Trop}(M/X)$. By the superlevel-flat criterion, it suffices to show that every set $\{P : u(P) \geq \lambda\}$ is a flat. In a finite simple modular matroid, a line-closed point set is a flat: choose an independent spanning set $a_1, \ldots, a_m$ in it; by induction all points of $H = \mathrm{cl}(a_1, \ldots, a_{m-1})$ are in it, and for $x \in \mathrm{cl}(H, a_m) \setminus H$, modularity makes the line $\mathrm{cl}(a_m, x)$ meet H in a point, so line-closure contains x.

Thus it is enough to prove line-closure. Let P_1, P_2 be in the superlevel set and $P_3 \leq P_1 \vee P_2$ a point. Put $H_i = P_i^\perp \subseteq Y$. Then $H_3 \supseteq H_1 \wedge H_2$. If $H_3 = H_1$ or H_2, there is nothing to prove. Otherwise H_1, H_2, H_3 are distinct hyperplanes of Y containing $Z = H_1 \wedge H_2$, where $\mathrm{rk}(Z) = k - 1$. Choose a basis B of Z and $e_i \in H_i \setminus Z$. In the rank-two interval $[Z, Y]$, the three points H_1, H_2, H_3 form a circuit; lifting it to M and applying Lemma 2 to p and B shows that $\min(u(P_1), u(P_2), u(P_3))$ has no unique finite minimizer. Since $u(P_1), u(P_2) \geq \lambda$, we get $u(P_3) \geq \lambda$. Hence $u \in \mathrm{Trop}(M/X)$.

Return to C. If $C \subseteq X$, all terms are ∞. Otherwise set $D = C \setminus X$. For $c \in D$,

$$p^\dagger_{A\cup c} = u(\mathrm{cl}_M(X \cup c)).$$

Every element of D is contained in a circuit of M/X inside D, because $c \in \mathrm{cl}_M(C - c)$ gives

$$\mathrm{rk}_{M/X}(D - c) = \mathrm{rk}_{M/X}(D).$$

If the displayed minimum over $c \in C \setminus A$ had a unique finite minimizer, its image point in M/X would be a unique finite minimizer on such a circuit, contradicting $u \in \mathrm{Trop}(M/X)$. The quotient criterion proves $\mathrm{Trop}(p^\dagger) \subseteq \mathrm{Trop}(M)$. $\square$

Lemma 7 (Involution and flat quotients)**.** *For every valuated quotient p, $(p^\dagger)^\dagger = p$. If p^F denotes the constant valuated matroid of $M/F \oplus U_{0,F}$, then*

$$(p^F)^\dagger = p^{F^\perp}.$$

Proof. Let p have rank k, and let $n = r - k$. If $I \in \binom{E}{k}$ is M-dependent, then both p_I and $(p^\dagger)^\dagger_I$ are ∞. If I is independent and $F = \mathrm{cl}_M(I)$, then

$$(p^\dagger)^\dagger_I = \overline{p^\dagger}(F^\perp).$$

Choose an M-basis J of $F^\perp$. Then

$$\overline{p^\dagger}(F^\perp) = p^\dagger_J = \bar{p}(\mathrm{cl}_M(J)^\perp) = \bar{p}((F^\perp)^\perp) = \bar{p}(F) = p_I.$$

For the flat quotient, let $f = \mathrm{rk}(F)$. A rank $r-f$ flat G satisfies $\bar{p}^F(G) < \infty$ iff $G \vee F = E$. Since $\mathrm{rk}(G) + \mathrm{rk}(F) = r$ and M is modular, this is equivalent to $G \wedge F = \varnothing$, i.e. G is a lattice complement of F. The anti-involution sends complements of F to complements of $F^\perp$, because it interchanges meets and joins and sends $\varnothing$ to E. Therefore the dagger support and values of p^F are exactly those of $p^{F^\perp}$. □

Lemma 8 (Annihilator formula)**.** *Let p be a valuated quotient of the simple matroid M. For $x \in \mathrm{Trop}(M)$, regard x as a rank-one valuated quotient and form $x^\dagger$. Then*

$$x \in \mathrm{Trop}(p^\dagger) \quad \Longleftrightarrow \quad \mathrm{Trop}(p) \subseteq \mathrm{Trop}(x^\dagger). \tag{$*$}$$

Consequently $p^\dagger$ depends only on the tropical linear space $\mathrm{Trop}(p)$.

Proof. Let $B = \{b_1, \dots, b_r\}$ be an M-basis and define its polar basis by

$$b_i^* = \mathrm{cl}_M(B - b_i)^\perp.$$

Because M is simple, b_i^* is a single element. For $S \subseteq \{1, \dots, r\}$,

$$\mathrm{cl}_M\{b_i^* : i \in S\} = \mathrm{cl}_M\{b_i : i \notin S\}^\perp,$$

since joins are changed to meets by $\perp$, and the flats spanned by subsets of B form a Boolean lattice. Hence $B^* = \{b_1^*, \dots, b_r^*\}$ is a basis and $(B^*)^* = B$.

Let p have rank k. Define a rank k valuated matroid μ_B on $\{1, \dots, r\}$ by

$$\mu_B(T) = p_{\{b_i : i \in T\}}.$$

It has rank k, because $\mathrm{supp}(p)$ is a quotient of M, so every M-basis spans $N = \mathrm{supp}(p)$. For $|S| = r - k$,

$$p^\dagger_{\{b_i^* : i \in S\}} = \bar{p}\big(\mathrm{cl}_M\{b_i^* : i \in S\}^\perp\big) = \bar{p}(\mathrm{cl}_M\{b_i : i \notin S\}) = \mu_B(\{1, \dots, r\} \setminus S).$$

Thus $p^\dagger|_{B^*} = \mu_B^*$.

We use the standard tropical annihilator identity: for a valuated matroid μ on Q and $u \in \mathbb{T}^Q$,

$$u \in \mathrm{Trop}(\mu^*) \quad \Longleftrightarrow \quad \mathrm{Trop}(\mu) \subseteq H_u,$$

where $H_u = \{y : \min_i(u_i + y_i) \text{ has no unique finite minimizer}\}$. Indeed, the circuits of μ^* are the cocircuits of μ, and $\mathrm{Trop}(\mu)$ is the tropical convex hull of the cocircuits, while H_u is tropically convex.

Assume first that $x \in \mathrm{Trop}(p^\dagger)$. Fix B and put $u_B = (x_{b_1^*}, \dots, x_{b_r^*})$. Since $p^\dagger|_{B^*} = \mu_B^*$, we have $u_B \in \mathrm{Trop}(\mu_B^*)$, hence $\mathrm{Trop}(\mu_B) \subseteq H_{u_B}$. If $y \in \mathrm{Trop}(p)$, then $y|_B \in \mathrm{Trop}(\mu_B)$, unless all its coordinates are ∞, in which case the following condition is automatic. Therefore

$$\min_i(y_{b_i} + x_{b_i^*})$$

has no unique finite minimizer. But for the rank $r-1$ quotient $x^\dagger$,

$$(x^\dagger)_{B-b_i} = x_{b_i^*}.$$

Thus the defining equation for $y \in \mathrm{Trop}(x^\dagger)$ holds on every M-basis B.

It remains only to check dependent r-sets K. If $\mathrm{rk}_M(K) \leq r-2$, all $(r-1)$-subsets $K-e$ are dependent and all terms are ∞. If $\mathrm{rk}_M(K) = r-1$, then K has a unique M-circuit D. The finite terms occur exactly for $e \in D$, and $\mathrm{cl}_M(K-e)$ is the same hyperplane for all such e; hence the equation reduces, up to adding a common constant, to the circuit equation for D, which holds because $y \in \mathrm{Trop}(M)$. Thus $y \in \mathrm{Trop}(x^\dagger)$, and $\mathrm{Trop}(p) \subseteq \mathrm{Trop}(x^\dagger)$.

Conversely assume $\mathrm{Trop}(p) \subseteq \mathrm{Trop}(x^\dagger)$. Fix B and u_B. By the annihilator identity, it suffices to show that each cocircuit of μ_B lies in H_{u_B}. Such a cocircuit has the form $i \mapsto p_{A\cup b_i}$, where $A = \{b_i : i \in S\}$ and $|S| = k-1$. It is the restriction to B of the valuated cocircuit $e \mapsto p_{A\cup e}$ of p. If this vector is identically ∞, there is nothing to prove; otherwise it is a point of $\mathrm{Trop}(p)$, hence of $\mathrm{Trop}(x^\dagger)$. Testing the defining equation of $\mathrm{Trop}(x^\dagger)$ on the basis B gives

$$\min_i (p_{A\cup b_i} + x_{b_i^*})$$

with no unique finite minimizer. Thus $u_B \in \mathrm{Trop}(\mu_B^*)$.

Since $B \mapsto B^*$ is a bijection on M-bases, the previous paragraph says that x satisfies the defining equations of $p^\dagger$ on every Boolean apartment coming from an M-basis. Let $J \in \binom{E}{n+1}$, where $n = \mathrm{rk}(p^\dagger)$. If J is independent, extend it to an M-basis and use the apartment calculation. If $\mathrm{rk}_M(J) \leq n-1$, all n-subsets $J-j$ are dependent. If $\mathrm{rk}_M(J) = n$, then J has a unique circuit D; the finite coefficients $p^\dagger_{J-j}$ occur exactly for $j \in D$, and for those j the flats $\mathrm{cl}_M(J-j)$ are equal. The equation is therefore the circuit equation for D, satisfied by $x \in \mathrm{Trop}(M)$. Hence $x \in \mathrm{Trop}(p^\dagger)$.

Thus $(*)$ holds, and the right-hand side depends only on $\mathrm{Trop}(p)$. Therefore so does $\mathrm{Trop}(p^\dagger)$. □

Proposition 9 (The theorem for simple matroids)**.** *Theorem 1 holds when M is simple.*

Proof. If $r = 0$, the assertion is immediate. Otherwise, for $X \in \mathrm{Dr}(M)$, choose a valuated quotient p with $X = \mathrm{Trop}(p)$ and define

$$\Phi(X) = \mathrm{Trop}(p^\dagger).$$

By Lemma 8, this is independent of the chosen p. Lemma 7 gives $\Phi^2 = \mathrm{id}$. If $X \subseteq Y$ and $x \in \Phi(Y)$, then by Lemma 8,

$$Y \subseteq \mathrm{Trop}(x^\dagger).$$

Hence $X \subseteq \mathrm{Trop}(x^\dagger)$, so $x \in \Phi(X)$. Thus $\Phi(Y) \subseteq \Phi(X)$, and Φ is order-reversing. Finally Lemma 7 gives

$$\Phi(\mathrm{Trop}(M/F \oplus U_{0,F})) = \mathrm{Trop}(M/F^\perp \oplus U_{0,F^\perp})$$

for every flat F. □

Proof of Theorem 1. Let $S = \mathrm{si}(M)$. By Lemma 3, $\Delta : \mathrm{Dr}(S) \to \mathrm{Dr}(M)$ is a poset isomorphism. The flat-lattice anti-involution on $\mathcal{L}(M) \cong \mathcal{L}(S)$ induces one on $\mathcal{L}(S)$. By Proposition 9, there is an order-reversing involution

$$\Phi_S : \mathrm{Dr}(S) \to \mathrm{Dr}(S)$$

extending the induced involution on flats. Define $\Phi_M : \mathrm{Dr}(M) \to \mathrm{Dr}(M)$ by

$$\Phi_M(\Delta(X)) = \Delta(\Phi_S(X)).$$

Because Δ is a poset isomorphism, Φ_M is an order-reversing involution.

If $F \in \mathcal{L}(M)$ corresponds to $\bar{F} \in \mathcal{L}(S)$, then $F^\perp$ corresponds to $\bar{F}^\perp$, and Lemma 3 gives

$$\mathrm{Trop}(M/F \oplus U_{0,F}) = \Delta\big(\mathrm{Trop}(S/\bar{F} \oplus U_{0,\bar{F}})\big).$$

Therefore

$$\begin{aligned} \Phi_M(\mathrm{Trop}(M/F \oplus U_{0,F})) &= \Delta\big(\Phi_S\big(\mathrm{Trop}(S/\bar{F} \oplus U_{0,\bar{F}})\big)\big) \\ &= \Delta\big(\mathrm{Trop}(S/\bar{F}^\perp \oplus U_{0,\bar{F}^\perp})\big) \\ &= \mathrm{Trop}(M/F^\perp \oplus U_{0,F^\perp}), \end{aligned}$$

as required. □

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